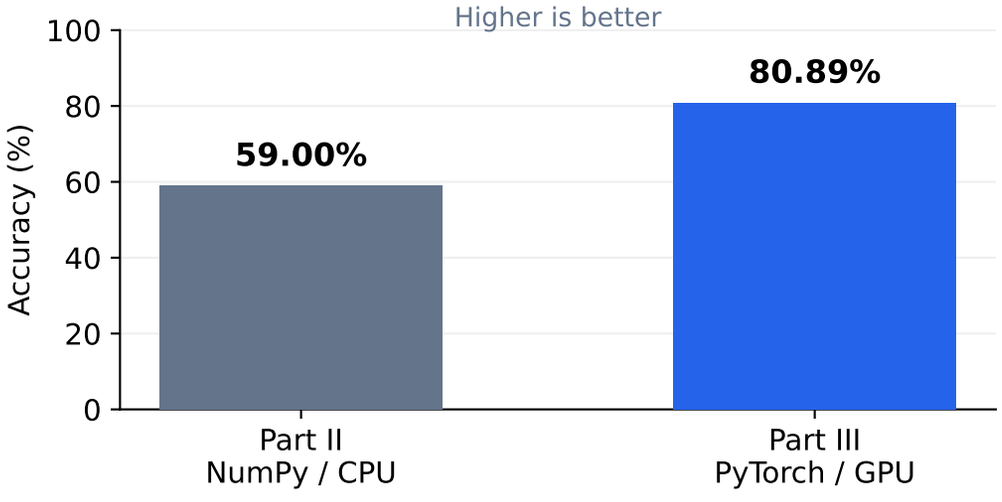
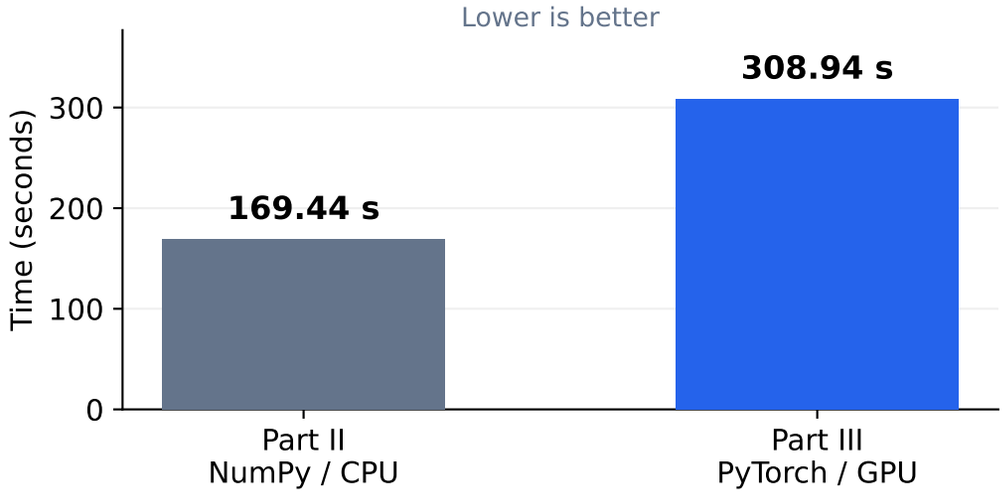
Cats vs. Dogs: NumPy and PyTorch Comparison

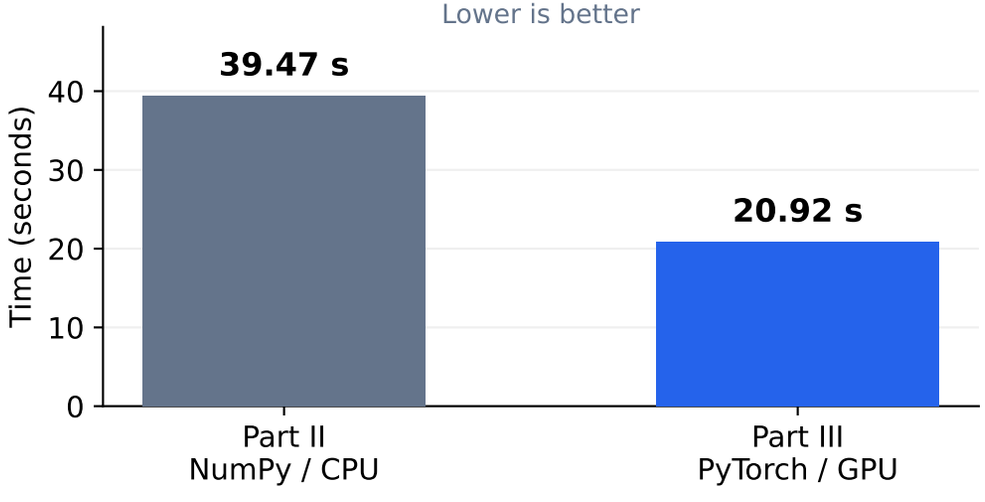
Test Accuracy



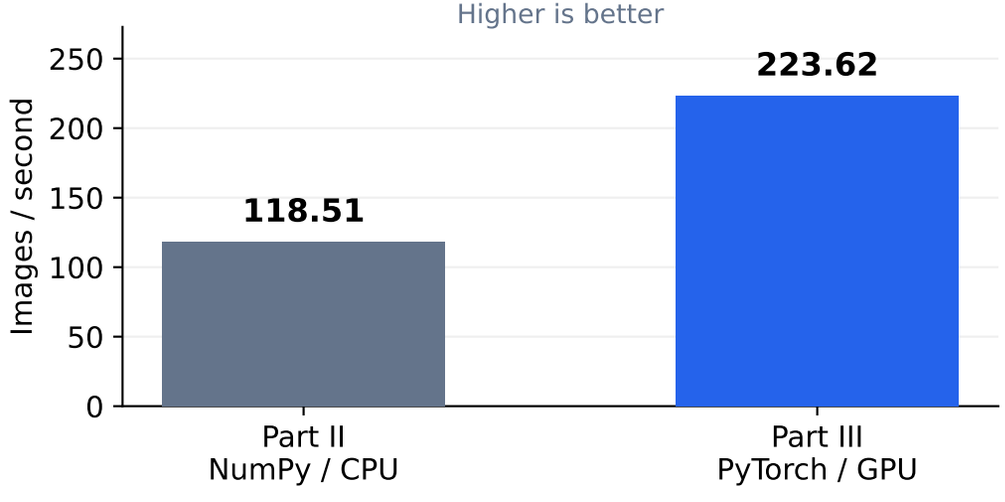
Total Training Time



End-to-End Test Time



End-to-End Test Throughput



Accuracy difference: +21.89 percentage points | Throughput ratio (III / II): 1.89x

Recorded runs use different hardware, architectures, and training methods.
Part II reuses fixed features; Part III trains the full CNN. This is not an isolated GPU speedup measurement.
Test timing includes image loading, preprocessing, and inference; Part III also includes CPU-to-GPU transfer.